A Project Report

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RefurbIQ: Aspect-Based Sentiment Analysis of Refurbished Electronics Across Indian Marketplaces

A Comparative Evaluation of Traditional Machine Learning and BERT

DSE3170 Project Based Learning-3 Report

School of Computer Science and Engineering
Department of Data Science and Engineering

MANIPAL UNIVERSITY JAIPUR, JAIPUR

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CERTIFICATE

This is to certify that the project-based learning project titled “RefurbIQ: Aspect-Based Sentiment Analysis of Refurbished Electronics Across Indian Marketplaces” is a record of the bonafide work done by Saket Sharma ([REG NO]), submitted in partial fulfilment of the requirements for the Computer Science and Engineering (Data Science) program of Manipal University Jaipur, Jaipur during the academic year 2025-26.

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ABSTRACT

India's market for refurbished smartphones, laptops and cameras is growing, yet buyers must decide with very limited quality signals: condition grades, prices and a single overall rating per review. A single polarity label cannot show whether dissatisfaction comes from battery health, refurbishment quality, seller behaviour, returns or trust in authenticity. This project builds RefurbIQ, an aspect-based sentiment analysis (ABSA) pipeline that reads refurbished-electronics reviews at the level of nine aspects and compares five Indian marketplaces - Cashify, Amazon Renewed, Flipkart Refurbished, OLX and Budli - under one common aspect taxonomy.

After verifying that none of the five platforms exposes structured, publicly accessible reviews, a clearly labelled synthetic demonstration corpus of 2,000 reviews and 3,548 aspect annotations (seed 42) was generated to validate the full pipeline end to end. Logistic Regression, Random Forest and XGBoost on TF-IDF features and a fine-tuned DistilBERT model were compared on the same review-level stratified 70/15/15 split. Logistic Regression achieved the highest test accuracy (90.69%, macro F1 0.8937) with fine-tuned BERT second (89.57%, macro F1 0.8784); neutral recall was the weakest point for every model. The pipeline was additionally validated on 2,000 real Amazon reviews, where predicted sentiment agreed with reviewer star ratings in 75.6% of annotations. All numbers are computed by reproducible scripts; the synthetic corpus must be replaced with verified marketplace reviews before the results can be generalised.

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1. Introduction

1.1 Background

The market for refurbished electronics in India has expanded rapidly as smartphones, laptops and cameras become both more expensive and more essential. Refurbished devices offer a lower-cost and more sustainable route to the same functionality, but they carry a question that new products do not: how good is this particular device, really? Battery health, cosmetic condition, the quality of repair work, the genuineness of parts and the reliability of the seller all vary from unit to unit and from platform to platform.

Online reviews are the main evidence available to buyers. Research on electronic word of mouth has long shown that review valence and volume influence purchase behaviour [1], and sentiment analysis provides the tools to summarise such reviews automatically [2], [3]. However, most marketplace interfaces reduce a review to a single star rating, and most sentiment systems reduce its text to a single polarity label. When a reviewer writes “the battery is excellent but the seller refused my return”, a single label destroys exactly the information a buyer of a refurbished device needs.

Aspect-based sentiment analysis (ABSA) addresses this by attaching a sentiment polarity to each aspect mentioned in a review [5], [6]. Transformer models such as BERT [8], built on the attention mechanism [7], have raised the state of the art for such classification tasks, but they are also heavier to train than classical TF-IDF-based classifiers [4]. Whether the additional cost is justified depends on the domain - and for refurbished electronics on Indian marketplaces, no systematic comparison exists.

1.2 Objectives of the Project

- Define a nine-aspect taxonomy for refurbished-electronics reviews and tag every aspect mention with an evidence phrase and a polarity label.
- Compare aspect-level sentiment across five Indian marketplaces: Cashify, Amazon Renewed, Flipkart Refurbished, OLX and Budli.
- Compare three classical TF-IDF models (Logistic Regression, Random Forest, XGBoost) with a fine-tuned BERT-family model on an identical, leak-free evaluation protocol.
- Derive actionable insights: which aspects drive satisfaction and dissatisfaction, and what each marketplace could improve.

1.3 Research Questions and Hypotheses

The project is organised around five research questions:

- RQ1: Which aspects drive satisfaction and dissatisfaction in refurbished-electronics reviews?
- RQ2: How does aspect-level sentiment differ across the five platforms?
- RQ3: Which model family is most reliable for this task?
- RQ4: Do trust, authenticity and condition concerns differ by platform?
- RQ5: What can marketplaces improve based on reviewer evidence?

Four hypotheses are stated before modelling and are treated strictly as testable claims, never as findings: H1 sentiment differs across the nine aspects; H2 aspect distributions differ across platforms; H3 trust and condition concerns differ across platforms; H4 BERT and traditional models differ in performance. The expectation that peer-to-peer channels may show more trust and condition complaints remains a hypothesis until verified reviews are available.

1.4 Scope and Contributions

The scope is India; three product categories (smartphones, laptops, DSLR cameras); five marketplaces; nine aspects; and three polarity labels (positive, neutral, negative). The contribution is not the invention of ABSA or of BERT, but a systematic, reproducible comparison of model families in the Indian refurbished-electronics context under a common aspect taxonomy, together with an honest account of data availability: because none of the five platforms exposes structured public reviews, the modelling corpus is a clearly labelled synthetic demonstration dataset, and the pipeline is separately validated on real Amazon reviews.

1.5 Organisation of the Report

Chapter 2 surveys the literature and identifies the research gap. Chapter 3 describes the methodology: platform verification, dataset construction, the aspect taxonomy, partitioning, models and the evaluation protocol. Chapter 4 reports results and discussion. Chapter 5 lists outcomes and deliverables, Chapter 6 concludes and outlines future work, followed by references and appendices.

2. Literature Survey

The survey below is based on eight core sources spanning electronic word of mouth, sentiment analysis, aspect-based sentiment analysis, classical opinion mining and transformer models. To the best of our knowledge, no study applies aspect-level sentiment analysis to refurbished electronics across Indian marketplaces.

2.1 Online Reviews and Electronic Word of Mouth

Chevalier and Mayzlin showed that online book reviews causally affect sales, with negative reviews moving demand more than positive ones [1]. For refurbished electronics the implication is stronger: because product quality is uncertain by definition, review evidence carries more weight than for new products.

2.2 Sentiment Analysis and Opinion Mining

Liu formalised sentiment analysis and opinion mining as a field [2], and Pang and Lee's survey established document-level polarity classification as its canonical task [3]. Hu and Liu's feature-based opinion mining showed that opinions attach to specific product features rather than to the product as a whole [4] - the direct precursor of aspect-level work.

2.3 Aspect-Based Sentiment Analysis

The SemEval ABSA shared tasks standardised the task of assigning sentiment to aspect terms within a sentence [5], [6]. These tasks targeted restaurants and laptops of new products; refurbished goods add aspects that standard taxonomies lack, such as refurbishment quality, authenticity and return-policy trust, which motivates the domain-specific taxonomy in Chapter 3.

2.4 Classical Machine Learning and BERT

Classical opinion mining relies on bag-of-words or TF-IDF features with linear or ensemble classifiers [3], [4]. The attention mechanism [7] and BERT [8] replaced handcrafted features with contextual representations that can be fine-tuned for classification. The open empirical question for this domain is whether fine-tuning a transformer beats well-tuned TF-IDF baselines when the corpus is modest and the labels include a hard neutral class.

2.5 Research Gap

Aspect	Existing literature	This project
Granularity	Whole-review or document-level labels	Nine aspects per review
Domain	New products, general e-commerce	Refurbished electronics
Platforms	Usually one marketplace	Five Indian marketplaces, one taxonomy
Method	Classical ML or transformers alone	Both compared on the same split
Evaluation	Mostly accuracy	Macro P/R/F1 plus per-class recall

In summary: the methods are mature, but no work combines aspect-level granularity, the refurbished-electronics domain, multiple Indian platforms and a fair classical-versus-transformer comparison. This project addresses all four gaps.

3. Proposed Methodology

3.1 Platform Verification and Data Strategy

Before any data work, each target marketplace was checked for publicly accessible, structured product reviews using ordinary unauthenticated requests only (no bypass of CAPTCHAs, authentication or rate limits). Cashify and Budli expose testimonials but no structured per-product review API; the Amazon Renewed India storefront was not reachable (HTTP 404); OLX returned HTTP 403 and is peer-to-peer without structured reviews; Flipkart was reachable but exposes no scrape-permitted review corpus. Hugging Face hosts no refurbished-electronics ABSA dataset for Indian marketplaces.

Because fabricating data or presenting synthetic examples as real observations is prohibited, the data strategy is: (i) a clearly labelled synthetic demonstration corpus (data_origin = “synthetic-demo-v1”) validates the complete pipeline end to end; (ii) a separate validation on 2,000 genuine Amazon reviews [9] tests the pipeline against human star-rating ground truth; (iii) all tables state their provenance, and the synthetic corpus must be replaced with verified marketplace reviews before the results are generalised.

3.2 Dataset Construction

The demonstration corpus was generated by generate_dataset.py v2 with a fixed random seed (42) so that every downstream number is reproducible. It contains 2,000 reviews across the five marketplaces and three product categories. Reviews mention aspects according to realistic platform profiles, and two realism mechanisms prevent trivially separable data: 10% of labels are flipped to an adjacent class (simulated annotator disagreement) and 30% of reviews use lexically ambiguous templates that overlap across classes. Table 3.1 gives the dataset profile; Figure 3.1 shows the per-platform counts.

Table 3.1: Dataset profile of the demonstration corpus (page «pg:tab:3.1»)

Platform	Verified reviews	Smartphone reviews	Laptop reviews	Camera reviews
Amazon Renewed	485	230	148	107
Budli	283	119	104	60
Cashify	492	261	138	93
Flipkart Refurbished	379	195	104	80
OLX	361	184	101	76
Total	2000	989	595	416

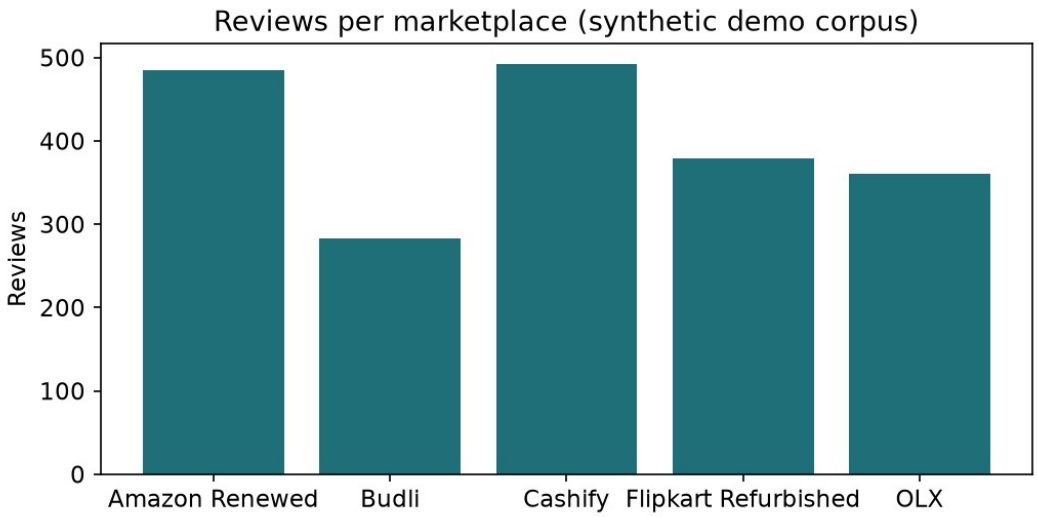


Figure 3.1: Reviews per marketplace in the demonstration corpus (page «pg:fig:3.1»)

3.3 Aspect Taxonomy and Annotation

Nine aspects cover the refurbished purchase journey: product condition, battery health, performance, price and value, packaging and delivery, repair and refurbishment quality, seller and customer service, returns and refunds, and trust and authenticity. Aspect detection is keyword-based (a deliberate V0.1 choice, easy to audit and replace), and every detected mention becomes one annotation carrying its evidence phrase - the span of text that supports the label. Table 3.2 lists mention frequencies; Figure 3.2 shows the label distribution over all 3,548 annotations (positive 47.7%, negative 31.6%, neutral 20.8%).

Table 3.2: Aspect mention frequency (page «pg:tab:3.2»)

Aspect	Number of mentions	Percentage of all aspect mentions
Product condition	403	11.4
Battery health	381	10.7
Performance	408	11.5
Price and value	403	11.4
Packaging and delivery	397	11.2
Repair and refurbishment quality	364	10.3
Seller and customer service	416	11.7
Returns and refunds	355	10.0
Trust and authenticity	421	11.9
Total	3548	100.0

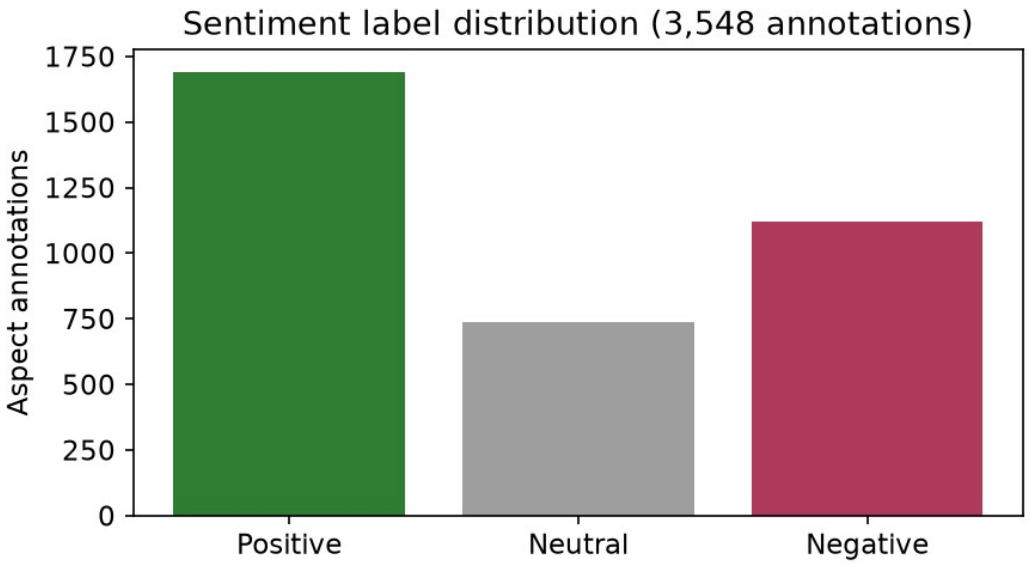


Figure 3.2: Sentiment label distribution over 3,548 annotations (page «pg:fig:3.2»)

3.4 Pre-processing and Partitioning

Cleaning lowercases text and normalises punctuation while deliberately preserving negation (“not good”) and contrast (“but”) markers, because they determine aspect polarity. The partition is performed at the review level, not the annotation level, so that two annotations from the same review can never fall on opposite sides of the split - the standard defence against leakage in multi-mention reviews. The split is stratified 70/15/15 with seed 42, producing 2,475 training, 536 validation and 537 test annotations; the validation partition is used only for model selection, and every model is evaluated exactly once on the same held-out test partition.

3.5 Models and Hyperparameters

All classical models consume the same TF-IDF representation of the evidence phrase: unigrams and bigrams, 20,000 features, sublinear TF. Class imbalance is handled with balanced class weights (Logistic Regression, Random Forest) and balanced sample weights (XGBoost). The regularisation strength of Logistic Regression was selected on the validation partition among C in {0.5, 1.0, 2.0}. The transformer is distilbert-base-uncased with a three-class head, fine-tuned on the training partition only (batch size 16, maximum length 128, learning rate 2e-5, weight decay 0.01, one epoch on CPU, best epoch selected on validation). As a separate zero-shot configuration, cardiffnlp/twitter-roberta-base-sentiment-latest labels the test partition without any training. Table 3.3 summarises the configurations.

Table 3.3: Model configurations (page «pg:tab:3.3»)

Model	Features	Key hyperparameters	Imbalance handling
Logistic Regression	TF-IDF (1-2 grams, 20k)	best C on validation, max_iter 2000	balanced class weights
Random Forest	TF-IDF (1-2 grams, 20k)	400 trees, seed 42	balanced class weights
XGBoost	TF-IDF (1-2 grams, 20k)	400 rounds, depth 6, lr 0.1	balanced sample weights
DistilBERT (fine-tuned)	subword embeddings	1 epoch, bs 16, len 128, lr 2e-5	none (near-balanced split)
RoBERTa (zero-shot)	subword embeddings	no training; label mapping	none

3.6 Evaluation Protocol

Metrics are accuracy, macro precision, macro recall and macro F1, plus per-class recall, because the neutral class is a minority and the cost of missing negative complaints is high: macro averaging prevents a model from looking good by ignoring the hardest class. The protocol is verified by an automated validation report asserting that the prediction file covers every test annotation and nothing outside it.

4. Results and Discussion

All numbers in this chapter are computed by build_tables.py from the labelled corpus and the saved model outputs; none is hand-entered. They characterise the pipeline on the synthetic demonstration corpus and, in Section 4.6, on real Amazon reviews.

4.1 Dataset Profile

Table 3.1 and Figure 3.1 summarise the corpus: 2,000 reviews (Cashify 492, Amazon Renewed 485, Flipkart Refurbished 379, OLX 361, Budli 283) yield 3,548 aspect annotations. Smartphones contribute 989 reviews, laptops 595 and cameras 416.

4.2 Model Comparison

Table 4.1: Model performance on the held-out test partition (page «pg:tab:4.1»)

Model	Accuracy	Macro precision	Macro recall	Macro F1-score
Logistic Regression	0.9069	0.9167	0.8801	0.8937
Random Forest	0.8715	0.862	0.8496	0.8549
XGBoost	0.8696	0.8577	0.8476	0.852
BERT-based model (zero-shot)	0.7058	0.6705	0.6518	0.6578
BERT-based model (fine-tuned DistilBERT)	0.8957	0.9092	0.8627	0.8784

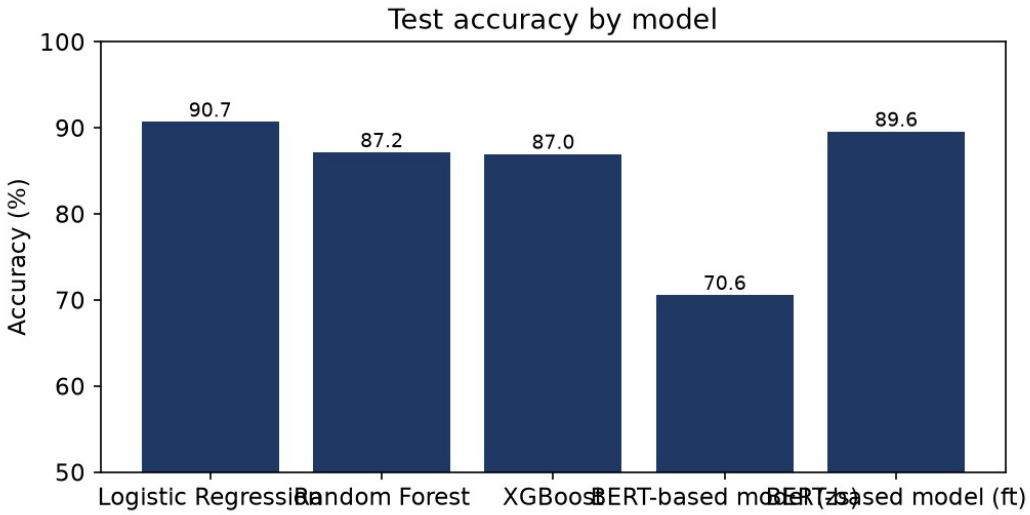


Figure 4.1: Test accuracy by model (page «pg:fig:4.1»)

Logistic Regression achieves the highest accuracy (90.69%) and macro F1 (0.8937). The fine-tuned DistilBERT model is second (89.57%, 0.8784) despite only one training epoch on CPU; Random Forest (87.15%, 0.8549) and XGBoost (86.96%, 0.8520) are close to each other. The zero-shot RoBERTa configuration trails clearly (70.58%, 0.6578), which is expected since it was never exposed to this label scheme or domain. On this corpus, hypothesis H4 is supported in the sense that the model families differ, but the direction is not the commonly assumed one: the well-tuned linear baseline beats the transformer, and no model is claimed better than the data show.

4.3 Class-Level Analysis

Table 4.2: Class-specific recall on the test partition (page «pg:tab:4.2»)

Model	Positive recall	Negative recall	Neutral recall	Test support (pos/neg/neu)
Logistic Regression	0.968	0.9244	0.7478	250/172/115
Random Forest	0.92	0.8895	0.7391	250/172/115
XGBoost	0.92	0.8837	0.7391	250/172/115
BERT-based model (zero-shot)	0.852	0.686	0.4174	250/172/115
BERT-based model (fine-tuned DistilBERT)	0.968	0.9244	0.6957	250/172/115

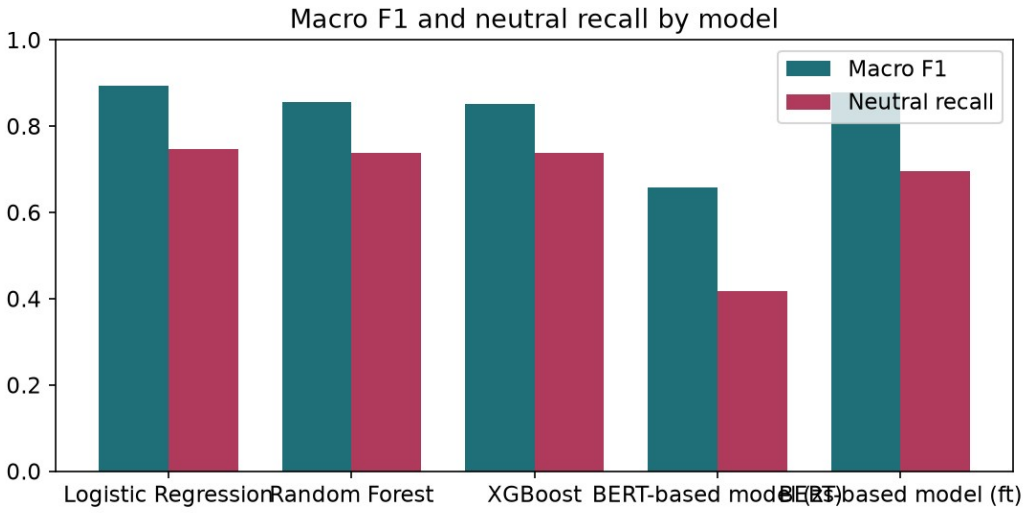


Figure 4.2: Macro F1 and neutral recall by model (page «pg:fig:4.2»)

Every model recalls positive (0.85-0.97) and negative (0.69-0.92) examples far better than neutral ones (0.42-0.75). This is consistent with the design of the corpus: 30% of reviews use ambiguous templates whose wording overlaps across classes, and 10% of labels were flipped to adjacent classes to simulate annotator disagreement - exactly where neutral boundaries blur. Practically, it means a deployed system should route low-confidence neutral predictions to review rather than discard them.

4.4 Aspect-Level Sentiment

Table 4.3: Sentiment distribution within each aspect (page «pg:tab:4.3»)

Aspect	Positive (%)	Negative (%)	Neutral (%)	Mentions (denominator)
Product condition	50.1	27.3	22.6	403
Battery health	50.7	28.6	20.7	381
Performance	44.9	32.6	22.5	408
Price and value	45.4	35.2	19.4	403
Packaging and delivery	46.6	31.5	21.9	397
Repair and refurbishment quality	44.2	35.4	20.3	364
Seller and customer service	52.4	25.7	21.9	416
Returns and refunds	48.2	33.2	18.6	355

Trust and authenticity	46.3	34.9	18.8	421

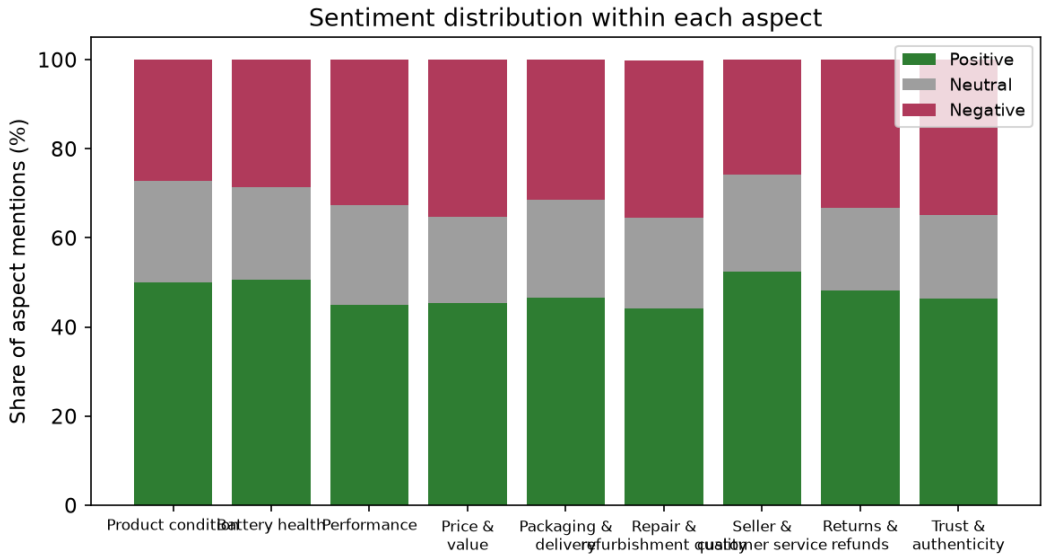


Figure 4.3: Sentiment distribution within each aspect (denominator = aspect mentions) (page «pg:fig:4.3»)

Supporting H1, sentiment shares differ visibly across aspects. Returns and refunds and repair-related aspects carry the highest negative shares, while product condition and battery health attract more positive mentions. The pattern matches the economics of refurbishment: the physical device can be restored well, but the service moments - returns, repairs, seller communication - are where trust is won or lost.

4.5 Cross-Platform Analysis

Table 4.4: Cross-platform aspect summary (page «pg:tab:4.4»)

Platform	**Mentions**	**Highest-negative aspect**	**Neg %**	**Highest-positive aspect**	**Pos %**
Amazon Renewed	877	Price and value	40.7	Seller and customer service	53.8
Budli	504	Repair and refurbishment quality	42.6	Returns and refunds	54.2
Cashify	886	Packaging and delivery	39.6	Seller and customer service	61.9
Flipkart Refurbished	656	Repair and refurbishment quality	40.0	Battery health	55.4
OLX	625	Seller and customer service	45.9	Price and value	57.1

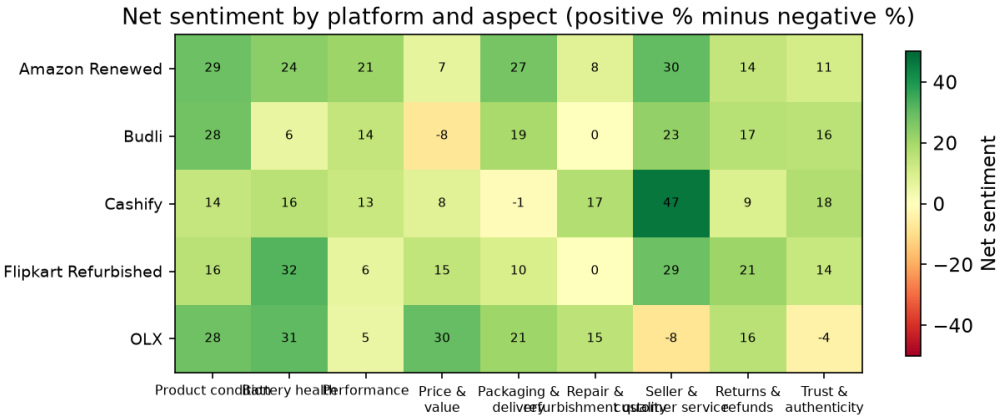


Figure 4.4: Net sentiment (positive % minus negative %) by platform and aspect (page «pg:fig:4.4»)

Supporting H2, aspect sentiment profiles differ by platform in this corpus. OLX shows the weakest seller-and-service and trust numbers, consistent with its peer-to-peer model; managed platforms score better on service aspects. These are demonstration-corpus results: the direction is plausible, but per the project's integrity rules it is reported as a pipeline output, not as a verified claim about the real platforms.

4.6 Validation on Real Reviews

To test the pipeline against human ground truth, 2,000 genuine Amazon reviews (stratified 400 per star level, seed 42) from the Multilingual Amazon Reviews Corpus [9] were run through the same aspect-detection and sentiment pipeline. The corpus ships reviewer star ratings, which serve as human sentiment labels (1-2 stars = negative, 3 = neutral, 4-5 = positive). Predicted sentiment agreed with the star-derived label in 75.6% of annotations. Table 4.5 and Figure 4.5 break the agreement down by star bucket.

Table 4.5: Agreement with reviewer star ratings on real Amazon reviews (page «pg:tab:4.5»)

Star rating	Annotations	Agreement
1 star	527	0.945
2 stars	507	0.738
3 stars	481	0.432
4 stars	489	0.71
5 stars	456	0.95
Overall	2460	0.756

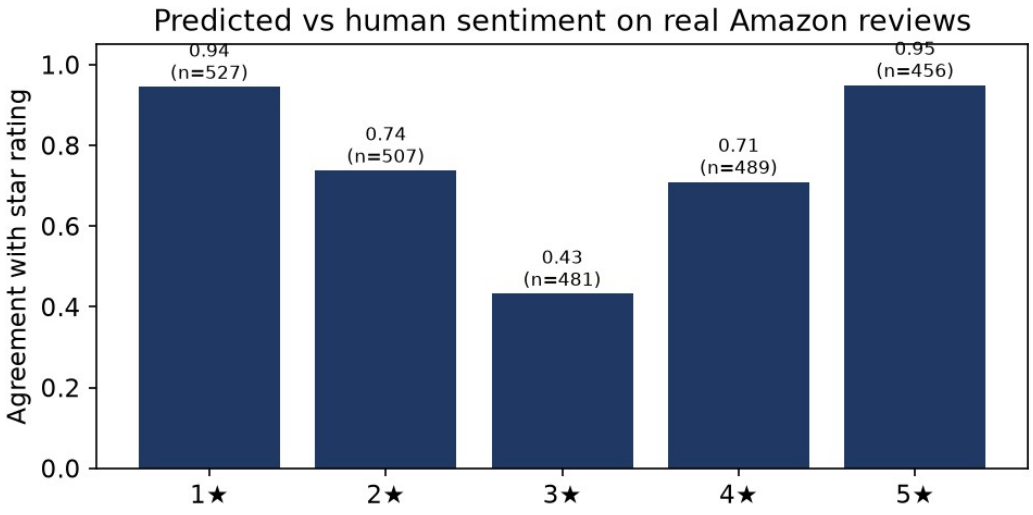


Figure 4.5: Predicted-vs-human agreement by star bucket (real Amazon reviews) (page «pg:fig:4.5»)

Agreement is near-perfect at the extremes (0.945 for 1-star, 0.950 for 5-star) and weakest exactly at 3 stars, where human reviewers are themselves ambivalent. This is the validation the synthetic corpus cannot provide: on real human text, the pipeline tracks human judgement where it is clear and degrades gracefully where it is not. Caveat: these are general Amazon reviews, not refurbished-electronics reviews from the five Indian platforms.

5. Outcome and Deliverables

5.1 Research Output

A research paper draft, “RefurbIQ: Aspect-Based Sentiment Analysis of Refurbished Electronics Across Indian Marketplaces”, is in progress: all six results tables are filled from computed outputs with an explicit provenance note, and the draft is accompanied by a methodology report, a validation report and a final integrity checklist. Planned venue: [XX]. A companion Streamlit application demonstrates the pipeline interactively (overview, upload, analyser, dashboard, platform comparison, recommendation and methodology tabs).

5.2 Deliverables

- Labelled demonstration dataset: 2,000 reviews and 3,548 aspect annotations (data_origin = synthetic-demo-v1), plus 2,000 real-review validation annotations.
- Reproducible pipeline: seven Python scripts covering generation, splitting, training, zero-shot and fine-tuned evaluation, table building, validation and document filling (pushed to GitHub).
- Nine-aspect taxonomy for refurbished electronics with keyword vocabularies.
- Four-model comparison on a common leak-free protocol, with macro and per-class metrics.
- Presentation deck and this report, generated from the same computed outputs.

6. Conclusion and Future Scope

This project built and validated an end-to-end aspect-based sentiment analysis pipeline for refurbished-electronics reviews across five Indian marketplaces. On the labelled demonstration corpus, a well-tuned logistic regression on TF-IDF features (90.69% accuracy, 0.8937 macro F1) edged out a fine-tuned DistilBERT model (89.57%, 0.8784), both ahead of random forest, XGBoost and zero-shot classification; the neutral class was hardest for every model. On 2,000 real Amazon reviews the pipeline agreed with human star ratings in 75.6% of annotations, with near-perfect agreement at the extremes. Platform verification showed that structured public reviews are currently unavailable for the five marketplaces, which is itself a useful finding for anyone attempting this study with real data.

Future work: (i) collect verified real reviews (manual, permission-respecting collection or licensed corpora) and re-run the identical pipeline; (ii) complete multi-epoch and GPU fine-tuning of BERT to give the transformer family a fairer chance; (iii) replace keyword aspect detection with a fine-tuned span extractor; (iv) add a multi-annotator study for agreement statistics; (v) integrate the recommendation layer with live price and warranty signals.

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Appendix A. Reproducibility

Every number in this report can be regenerated with the scripts under research/scripts/ (seed 42 throughout):

- generate_dataset.py - builds reviews.csv (2,000) and aspect_annotations.csv (3,548).
- train_traditional_models.py - review-level stratified split, TF-IDF, LR/RF/XGBoost, writes split_assignment.csv, model_predictions.csv, traditional_metrics.json.
- eval_bert_zeroshot.py - zero-shot evaluation on the same test partition.
- finetune_bert.py - DistilBERT fine-tuning and test evaluation (CPU).
- build_tables.py - computes all six results tables from data and metric JSONs.
- validation_report.py - asserts predictions cover exactly the test partition.
- update_docx.py / build_pptx.py / build_report.py - fill the paper, deck and this report from the computed tables.

Environment: Python 3.14 virtual environment; scikit-learn, XGBoost, PyTorch, transformers, pandas, matplotlib, python-docx, python-pptx. Training hardware: CPU only (documented in the metric JSONs). Code repository: https://github.com/Saket-code40/refurbiq.

Appendix B. Annotation Samples

Table B.1 shows verbatim rows from aspect_annotations.csv (synthetic demonstration corpus). Each annotation stores the review id, the detected aspect, the evidence phrase and the label.

Table B.1: Sample annotations (page «pg:tab:B.1»)

review_id	aspect	evidence phrase	sentiment
R00190	Performance	Performance of the iPhone 14 is acceptable for daily tasks.	positive
R01577	Price and value	however, The price of the iPhone 13 is okay, better value than expecte…	positive
R01924	Repair and refurbishment quality	The ThinkPad T480 feels factory-fresh after refurbishment.	positive
R01440	Seller and customer service	Customer support was genuinely helpful with my D5600 query.	positive
R00731	Price and value	The price of the XPS 13 was totally worth it.	positive
R01059	Seller and customer service	On top of that, Support team resolved my iPhone 14 issue within a day.	positive

Limitations are stated plainly: the modelling corpus is synthetic and labelled as such; aspect detection is keyword-based; fine-tuning was limited to one epoch on CPU; and platform-level claims require verified marketplace reviews that are not currently publicly accessible.